

ON THE DISTRIBUTION OF THE SAMPLE COVARIANCE MATRIX FROM A MIXTURE OF NORMAL DENSITIES

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SUMMARY

In this paper we consider sampling from a mixture of r -dimensional normal populations. It is shown that the distribution of sample means and sample covariance matrix is a linear combination of products of multivariate normal and noncentral Wishart densities with weights given by the multinomial distribution. Next we consider a multivariate random effects model and assume that the random variables follow a mixture of two multivariate normal distributions. It is shown that the joint distribution of sum of square matrices is a linear combination of products of noncentral Wishart densities with weights given by binomial and multidimensional hypergeometric distributions.

1. Introduction

In making statistical inferences about parametric multivariate analysis, it is conveniently assumed that the universe Π is a normal population (see Anderson (1958) for example). Few results are available if Π is not normal, aside from some asymptotic results such as given in Ito (1969). One difficulty appears to be that it is rather hard to derive or approximate distributions of sample covariances from general non-normal populations. By assuming that Π is a mixture of normal distributions, we derive the distributions of sample covariance matrices and of the between/within sums of squares matrices in a multivariate random effects model. These provide extensions of results given in Hyrenius (1950, 1952) and Subrahmaniam (1972). It might be of interest to note, as pointed out by Hyrenius (1950), that mixtures of normals are fairly frequent in medical and biological experiments. Also, as shown in Tan and Chang (1972) and Tan (1975), quantitative genetic models are basically mixtures of normal densities.

In Section 3, we derive the distribution of the sample covariance matrix from a mixture of m r -dimensional normal distributions. By assuming that the random variables are random samples from mixtures of two normal densities, we derive in Section 4 the

distributions of sum of square matrices in a multivariate random effects model. The results derived in this paper will be utilised in another paper to study the effect of departure from normality on classical inference procedures.

2. Some preliminaries and lemmas

Working with mixtures of densities, one is usually confronted with permutation operations (see Behboodian (1972), Subrahmaniam (1972), among others). For making extensions possible to more complicated models, we present in this paper some notation which would reduce the permutation operations into algebraic operations. We let δ_i and δ_{ij} be nonnegative integers taking values 0 or 1 only, and a_i and t_i , $i = 1, \dots, k$, nonnegative

integers. Define $W_a(\delta_i) = \{(\delta_i, i = 1, \dots, k) | \sum_{i=1}^k \delta_i = a\}$,

$$W_a(\delta_{ij}) = \{(\delta_{ij}, i = 1, \dots, k, j = 1, \dots, n) | \sum_{i=1}^k \sum_{j=1}^n \delta_{ij} = a\},$$

$$W_a(a_i) = \{(a_i, i = 1, \dots, k) | \sum_{i=1}^k a_i = a\} \text{ and } W_t(t_i) =$$

$$\{(t_i, i = 1, \dots, k) | \sum_{i=1}^k t_i = t\},$$

where $W_s(\delta_i)$ is the set of all $(\delta_1, \dots, \delta_k)$ satisfying $\sum_{i=1}^k \delta_i = s$, etc. Further, we denote by

$$H(\delta_{ij}; a_u) = \{(\delta_{ij}, i = 1, \dots, m, j = 1, \dots, n) | \sum_{i=1}^m \delta_{ij} = 1, j = 1, \dots, n \text{ and}$$

$$\sum_{j=1}^n \delta_{ij} = a_i, i = 1, \dots, m\} \text{ the set of all } (\delta_{11}, \delta_{12}, \dots, \delta_{mn}) \text{ satisfying the conditions}$$

$$\sum_{i=1}^m \delta_{ij} = 1, j = 1, \dots, n \text{ and } \sum_{j=1}^n \delta_{ij} = a_i, i = 1, \dots, m, \text{ and let } \sum_S \text{ denote the summation}$$

over the elements of S . Then the following three lemmas, whose proofs are straightforward, are obtained.

Lemma 1

$$(2.1) \prod_{j=1}^n \left\{ \sum_{i=1}^m b_{ij} \right\} = \sum_{W_n(a_u)} H(\delta_{ij}; a_u) \left\{ \prod_{i=1}^m \prod_{j=1}^n b_{ij}^{\delta_{ij}} \right\},$$

Lemma 2

Let $g(\delta_{ij}; i = 1, \dots, k, j = 1, \dots, n)$ be a function of $a_i = \sum_{j=1}^n \delta_{ij}$, $i = 1, \dots, k$, then

$$(2.2) \sum_{W_k(\delta_{ij})} g(\delta_{ij}; i = 1, \dots, k, j = 1, \dots, n) = \binom{kn}{a} \sum_{W_a(a_u)} h(n, a_1, \dots, a_k) \times g(a_1, \dots, a_k)$$

where $h(n, a_1, \dots, a_k) = \frac{k!}{\prod_{j=1}^k a_j!} \binom{n}{a_1, \dots, a_k} / \binom{kn}{a}$, $\sum_{j=1}^k a_j = a$.

Lemma 3

Let $\psi(\delta, t) = h(t_1, \dots, t_k) + c_0 \sum_{i=1}^k \delta_i t_i$ and $\eta(t) = h(t_1, \dots, t_k) + c_0 \sum_{i=1}^s t_i$. Then, for any function $g(x)$ of x ,

$$(2.3) \quad \sum_{s=1}^k W_s(\delta_i) \sum_{t=1}^k g(\psi(\delta, t)) = \binom{k}{s} W_s(t_i) \sum_{t=1}^k g(\eta(t)),$$

The following lemma which is well known in multivariate analysis will be used in deriving our main results. This is known as the multivariate Cochran theorem (Anderson (1958)).

Lemma 4

Let X be a $n \times r$ ($n \geq r$) matrix of random variables having density

$$f(X) = \left(\frac{1}{\sqrt{2\pi}} \right)^{nr} |V|^{-\frac{n}{2}} \exp \left\{ -\frac{1}{2} \text{tr} V^{-1} (X-D)' (X-D) \right\}.$$

Let A_j , $j = 1, \dots, k$ be symmetric $n \times n$ matrices satisfying $A_j^2 = A_j$ and $A_j A_{j'} = 0$ for $j \neq j'$. Let $a \neq 0$ be $n \times 1$ such that $A_j a = 0$, $j = 1, \dots, k$. Then, $z' = a' X$ and $Q_j = X' A_j X$, $j = 1, \dots, k$ are independently distributed as $z \sim N(D'a, (a'a)V)$, a normal density with expectation $D'a$ and covariance matrix $(a'a)V$, and $Q_j \sim W_p(\rho_j, V, N_j)$, a noncentral Wishart density with degrees of freedom $\rho_j = \text{rank } A_j$, parameter V and noncentrality parameter $N_j = D' A_j D V^{-1}$.

3. The distribution of sample covariance matrix from mixtures of normals

Let $x_1, \dots, x_n$ be a random sample from a mixture of m r -dimensional normal universes, with density

$$\phi(x) = \sum_{i=1}^m p_i N(x; \mu_i, V). \text{ Put } \bar{x} = \frac{1}{n} \sum_{j=1}^n x_j \text{ and } S = \sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})'.$$

With $\delta_{ij} \geq 0$ being nonnegative integers taking values 0 and 1 and satisfying

$\sum_{i=1}^m \delta_{ij} = 1$, $j = 1, \dots, n$ and $\sum_{j=1}^n \delta_{ij} = a_i$, $i = 1, \dots, m$, we have, after some manipulation

$$(3.1) \quad \prod_{i=1}^m \prod_{j=1}^n [p_i N(x_j; \mu_i, V)]^{\delta_{ij}} = \left(\frac{1}{\sqrt{2\pi}} \right)^{rn} \left(\prod_{i=1}^m p_i^{a_i} \right) |V|^{-\frac{1}{2}n} \exp \left\{ -\frac{1}{2} \text{tr} V^{-1} \left[\sum_{j=1}^n \right. \right.$$

$$\left. \left. (x_j - \bar{\mu} - \lambda_j)(x_j - \bar{\mu} - \lambda_j)'\right] \right\}$$

$$= \left(\frac{1}{\sqrt{2\pi}} \right)^{rn} \left(\prod_{i=1}^m p_i^{a_i} \right) |V|^{-\frac{1}{2}n} \exp \left\{ -\frac{1}{2} \text{tr} V^{-1} [X-U-\Lambda] [X-U-\Lambda]' \right\}$$

where $\bar{\mu} = \frac{1}{m} \sum_{i=1}^m \mu_i$, $\lambda_j = \sum_{i=1}^m \delta_{ij}(\mu_i - \bar{\mu})$, $X = (x_1, \dots, x_n)$, $U = (\bar{\mu}, \dots, \bar{\mu})$ and

$\Lambda = (\lambda_1, \dots, \lambda_n)$. By Lemma 1, the joint distribution of $x_1, \dots, x_n$ is then given by

$$(3.2) \quad \prod_{j=1}^n (\phi(x_j)) = \sum_{u=1}^n \left(\prod_{v=1}^m p_v^{a_v} \right) H(\delta_{ij} a_u) \left(\frac{1}{\sqrt{2\pi}} \right)^{rn} |V|^{-\frac{1}{2}n} \exp \left\{ -\frac{1}{2} \text{tr} V^{-1} (X-U-\Lambda) \times (X-U-\Lambda)' \right\}.$$

Writing $\bar{X} = \frac{1}{n} \sum_{j=1}^n x_j = X \left(\frac{1}{n} 1_n \right) = Xa$ and $S = \sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})'$

$= X' \left(I_n - \frac{1}{n} J_n \right) X = X' A X$, where 1_n is a $n \times 1$ column of 1's, $a = \frac{1}{n} 1_n$ and J_n a $n \times n$ matrix of 1's, it is obvious that $A^2 = A$ and $Aa = 0$.

Using (3.2) and applying Lemma 4, we have then the following theorem for the joint distribution of $\bar{X}$ and S :

Theorem 3.1

The joint density function $h(\bar{X}, S)$ of $\bar{X}$ and S is given by:

$$(3.3) \quad h(\bar{X}, S) = \sum_{u=1}^n \binom{n}{a_1, \dots, a_{m-1}} \left\{ \prod_{i=1}^m p_i^{a_i} \right\} \phi \left(\bar{X}, \bar{\mu}_a \frac{1}{a' n} V \right) G_f(S; V, N_a),$$

where $\binom{n}{a_1, \dots, a_{m-1}} = (n!) / \left\{ \prod_{i=1}^m a_i! \right\}$, $f = n-1$, $\phi \left(\bar{X}, \bar{\mu}_a \frac{1}{a' n} V \right)$ is a normal density with mean $\bar{\mu}_a = \frac{1}{n} \sum_{u=1}^m a_u \mu_u$ and covariance matrix $\frac{1}{n} V$, and $G_f(S; V, N_a)$ a noncentral Wishart density with d.f. = f , parameter matrix V and noncentrality matrix

$$N_a = \left[\sum_{u=1}^m a_u (\mu_u - \bar{\mu})(\mu_u - \bar{\mu})' \right] V^{-1}.$$

Theorem 3.1 extends some results given in Hyrenius (1950, 1952).

From (3.3), one sees immediately that the marginal distributions $h_1(\bar{X})$ of $\bar{X}$ and $h_2(S)$ of S are given respectively by

$$(3.4) \quad h_1(\bar{X}) = \sum_{u=1}^n \binom{n}{a_1, \dots, a_{m-1}} \left(\prod_{v=1}^m p_v^{a_v} \right) \phi \left(\bar{X}, \bar{\mu}_a \frac{1}{a' n} V \right),$$

and

$$(3.5) \quad h_2(S) = \sum_{n(a_u)} \binom{n}{a_1, \dots, a_{m-1}} \left(\prod_{v=1}^m p_v^{a_v} \right) G_F(S; V, N_a)$$

When $r=1$, (3.5) reduces to a result given in Hyrenius (1950). When $m=1$, (3.5) appears to be the multivariate analog of a result given in Subrahmaniam (1972).

4. Distributions of sum of square matrices in a multivariate random effects model

Consider the multivariate one-way random effect model

$$U_{ij} = \mu + \alpha_i + e_{ij}, \quad i = 1, \dots, k, \quad j = 1, \dots, n,$$

where $\alpha_1, \dots, \alpha_k$ is a random samples from a r -dimensional universe Π_α independent of the e_{ij} 's and e_{ij} , $i = 1, \dots, k$, $j = 1, \dots, n$, is a random sample from a r -dimensional universe Π_e . We assume that Π_α and Π_e are mixtures of two normals, with densities given respectively be

$$\phi_1(\alpha) = p_1 N(\alpha; \Delta_{11}, V_{11}) + q_1 N(\alpha; \Delta_{12}, V_{12}) \text{ and}$$

$$\phi_2(e) = p_2 N(e; \Delta_{21}, V_{21}) + q_2 N(e; \Delta_{22}, V_{22}), \text{ where } 0 \leq p_i \leq 1, \quad q_i = 1 - p_i, \quad i = 1, 2$$

and $N(X; \mu, V)$ a normal density with mean μ and covariance matrix V . For obtaining manageable results and for complying with

$$E\alpha_i = Ee_{ij} = 0, \text{ it is assumed that}$$

$$\Delta_{u1} = q_u \Delta_u, \Delta_{u2} = -p_u \Delta_u \text{ and}$$

$$V_{u1} = V_{u2} = V_u, \quad u = 1, 2.$$

Put

$$\bar{Y}_{i\cdot} = \frac{1}{n} \sum_{j=1}^n Y_{ij},$$

$$\bar{Y}_{\cdot\cdot} = \frac{1}{k} \sum_{i=1}^k \bar{Y}_{i\cdot},$$

$$S_b = n \sum_{i=1}^k (\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})(\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})'$$

and $S_w = \sum_{i=1}^k \sum_{j=1}^n (Y_{ij} - \bar{Y}_{i\cdot})(Y_{ij} - \bar{Y}_{i\cdot})'$. Under the above assumptions we now derive

the joint distribution of S_b and S_w . Now, from (3.2) of Section 3, we have for the joint density of $\alpha = (\alpha_1, \dots, \alpha_k)$:

$$(4.1) \quad f_1(\alpha) = \sum_{s=0}^k p_1 q_1^{k-s} \sum_{W_s(\delta_i)} \left(\frac{1}{\sqrt{2\pi}} \right)^{rk} |V_1|^{-\frac{k}{2}} \exp \left\{ -\frac{1}{2} \text{tr } V_1^{-1} (\alpha - U_\alpha)(\alpha - U_\alpha)' \right\},$$

where $U_\alpha = \Delta \delta' - p_1 \Delta_1 l'_k$ with $\delta' = (\delta_1, \dots, \delta_k)$.

Putting $E = (E_1, \dots, E_k)$, $E_i = (e_{i1}, \dots, e_{in})$, $U_e = \Delta_2 \delta' - p_2 \Delta_2 l'_{nk}$, where $\delta' = (\delta'_1, \dots, \delta'_k)$ with $\delta'_i = (\delta'_{i1}, \dots, \delta'_{in})$, $i = 1, \dots, k$, we have similarly for the joint distribution of e_{ij} , $i = 1, \dots, k$, $j = 1, \dots, n$:

$$(4.2) \quad f_2(E) = \sum_{t=0}^{kn} p_2 q_2^{kn-t} \sum_{W_t(\delta_{ij})} \left(\frac{1}{\sqrt{2\pi}} \right)^{knr} |V_2|^{-\frac{1}{2}kn} \exp \left\{ -\frac{1}{2} \text{tr } V_2^{-1} (E - U_e)(E - U_e)' \right\},$$

Combining (4.1) and (4.2) with the model, it can readily be shown that the likelihood function of $(\mu, p_1, p_2, \Delta_1, \Delta_2, V_1, V_2)$ is:

$$(4.3) \quad L(\mu, p_1, p_2, \Delta_1, \Delta_2, V_1, V_2 | Y) = \sum_{s=0}^k \sum_{t=0}^{kn} p_1^s q_1^{k-s} p_2^t q_2^{kn-t} \\ \times \sum_{W_s(\delta_i)} \sum_{W_t(\delta_{ij})} \left(\frac{1}{\sqrt{2\pi}} \right)^{knr} |V_2|^{-\frac{1}{2}k(n-1)} \times |V_2 + nV_1|^{-\frac{1}{2}k} \\ \times \exp \left\{ -\frac{1}{2} \sum_{i=1}^k \sum_{j=1}^n \text{tr } V_2^{-1} [(Y_{ij} - \bar{Y}_{i.}) - (c_{ij} - \bar{c}_{i.})][(Y_{ij} - \bar{Y}_{i.}) - (c_{ij} - \bar{c}_{i.})]' \right. \\ \left. - \frac{n}{2} \text{tr}(V_2 + nV_1)^{-1} \left[\sum_{i=1}^k (\bar{Y}_{i.} - \bar{Y}_{..} - \bar{c}_{i.} + \bar{c}_{..}) (\bar{Y}_{i.} - \bar{Y}_{..} - \bar{c}_{i.} + \bar{c}_{..})' \right. \right. \\ \left. \left. + k(\bar{Y}_{..} - \bar{c}_{..})(\bar{Y}_{..} - \bar{c}_{..})' \right] \right\};$$

where $c_{ij} = \mu - p_1 \Delta_1 - p_2 \Delta_2 + \delta_i \Delta_1 + \delta_{ij} \Delta_2$, $\bar{c}_{i.} = \frac{1}{n} \sum_{j=1}^n c_{ij}$ and $\bar{c}_{..} = \frac{1}{k} \sum_{i=1}^k \bar{c}_{i.}$.

One has $c_{ij} - c_{i.} = \Delta_2(\delta_{ij} - \frac{t_j}{n})$ and

$$\bar{c}_{i.} - c_{..} = \Delta_1(\delta_i - \frac{s}{k}) + \Delta_2 \frac{1}{nk} (kt_i - t).$$

Putting $\Delta_{ij} = \Delta_i \Delta_j'$ and noting that $\sum_{j=1}^n \delta_{ij} = t_i$ and $\sum_{i=1}^k \delta_i = s$, we have:

$$N_e = \sum_{i=1}^k \sum_{j=1}^n (c_{ij} - \bar{c}_{i.})(c_{ij} - \bar{c}_{i.})' V_2^{-1} \\ = \Delta_{22} V_2^{-1} \left\{ \sum_{i=1}^k \sum_{j=1}^n [\delta_{ij} - \frac{2}{n} \delta_{ij} t_i + (\frac{t_i}{n})^2] \right\}$$

$$\begin{aligned}
&= \Delta_{22} V_2^{-1} \sum_{i=1}^k \frac{t_i(n-t_i)}{n} \quad \text{and} \\
N_{\alpha}^* &= n \sum_{i=1}^k (\bar{c}_{i\cdot} - \bar{c}..)(\bar{c}_{i\cdot} - \bar{c}..) (V_2 + nV_1)^{-1} \\
&= n \{ \Delta_{11} \sum_{i=1}^k [\delta_i - 2\delta_{i\frac{s}{k}} + (\frac{s}{k})^2] + (\Delta_{12} + \Delta_{21}) \frac{1}{n} \sum_{i=1}^k (\delta_i - \frac{s}{k})(t_i - \frac{t}{n}) \\
&\quad + \Delta_{22} \frac{1}{n^2} \sum_{i=1}^k (t_i - \frac{t}{n})^2 \} (V_2 + nV_1)^{-1} \\
&= \{ \Delta_{11} (\frac{ns(k-s)}{k}) + (\Delta_{12} + \Delta_{21}) \sum_{i=1}^k t_i (\delta_i - \frac{s}{k}) \\
&\quad + \Delta_{22} \frac{1}{n} \sum_{i=1}^k (t_i - \frac{t}{n})^2 \} (V_2 + nV_1)^{-1}.
\end{aligned}$$

$$\begin{aligned}
\text{Now, } S_b &= n \sum_{i=1}^k (\bar{Y}_{i\cdot} - \bar{Y}..)(\bar{Y}_{i\cdot} - \bar{Y}..) = Y' [I_k - \frac{1}{k} J_1] \otimes \frac{1}{n} J_n Y \\
&= Y' A_1 Y \text{ and } S_w = \sum_{i=1}^k \sum_{j=1}^n (Y_{ij} - \bar{Y}_{i\cdot})(Y_{ij} - \bar{Y}_{i\cdot})' = Y' [I_k \otimes (I_n - \frac{1}{n} J_n)] Y \\
&= Y' A_2 Y.
\end{aligned}$$

Obviously, $A_1^2 = A_1$, and $A_1 A_2 = 0$. Applying Lemma 4 to (4.3) and noting Lemmas 2 and 3, we have immediately the following result for the joint distribution S_b and S_w :

Theorem (4.1)

The joint density $g(S_b, S_w)$ of S_b and S_w is given by:

$$(4.5) \quad g(S_b, S_w) = \sum_{s=0}^k \sum_{t=0}^{kn} \binom{kn}{s} \binom{kn}{t} p_1^s p_1^t q_1^{k-s} q_2^{kn-t} \times$$

$$W_{t_i}^{\sum} h(n; t_1, \dots, t_k) G_{fb}(s_b; V_2 + nV_1, N_{\alpha}) G_{fe}(S_w; V_2, N_e)$$

where $f_b = k-1$, $f_e = k(n-1)$, $N_e = \Delta_{22} V_2^{-1} \sum_{i=1}^k \frac{t_i(n-t_i)}{n}$ and

$$N_{\alpha} = [\Delta_{11} (\frac{ns(k-s)}{k}) + (\Delta_{12} + \Delta_{21}) (\sum_{i=1}^s t_i - \frac{st}{k}) + \Delta_{22} \frac{1}{n} \sum_{i=1}^k (t_i - \frac{t}{n})^2] (V_2 + nV_1)^{-1}.$$

If $\Delta_1 = 0$ (or Π_{α} normal), then $N_{\alpha} = \Delta_{22} (V_2 + nV_1)^{-1} \frac{1}{n} \sum_{i=1}^k (t_i - \frac{1}{k} t)^2 = N_{\alpha 1}$ and

$N_e = \Delta_{22} V_2^{-1} \sum_{i=1}^k \frac{1}{n} t_i (n-t_i)$ are independent of δ_i , $i = 1, \dots, k$.

If $\Delta_2 = 0$ (or Π_e normal), then $N_e = 0$ and $N_\alpha = \Delta_{11} \frac{s(k-s)}{k} = N_{\alpha 2}$ are independent of $\delta_j, j = 1, \dots, k$ and $t_j, j = 1, \dots, k$.

Hence,

$$(4.6) \quad g(S_b, S_w) = \left\{ \sum_{s=0}^k \binom{k}{s} p_{11}^{s, k-s} G_{fb}(S_b, V_e + nV_1, N_{\alpha 2}) \right\} G_{fe}(S_w, V_e),$$

where $G_{fe}(S_w, V_e)$ is central Wishart with d.f. $= k(n-1)$ and parameter matrix V_e . Notice also that S_b and S_w are independent in this case.

In Theorem 4.1, if $r = 1$ so that $V_1 = \sigma_1^2$ and $V_2 = \sigma_2^2$, then the joint density of

$$X_1 = \frac{S_b}{\sigma_2^2 + n\sigma_1^2} \text{ and } X_2 = \frac{S_w}{\sigma_2^2} \text{ is:}$$

$$(4.7) \quad g(S_b, S_w) = \sum_{s=0}^k \sum_{t=0}^{kn} \binom{k}{s} \binom{kn}{t} p_\alpha^s p_t^{k-s} q_\alpha^{kn-t}$$

$$\sum_{t_1} h(n; t_1, \dots, t_k) p(S_b x_{k-1}^2, \eta_{1st}^2) p(S_w x_{k(n-1)}^2, \eta_{2t}^2),$$

$$\text{where } \eta_{1st}^2 = \frac{1}{(\sigma_2^2 + n\sigma_\alpha^2)} \left\{ n\Delta_1^2 \frac{s(k-s)}{k} + 2\Delta_1\Delta_2 \left(\sum_{i=1}^s t_i - \frac{st}{k} \right) \right.$$

$$\left. + \Delta_2^2 \frac{1}{n} \sum_{i=1}^k \left(t_i - \frac{1}{n}t \right)^2 \right\}, \eta_{2t}^2 = \frac{1}{n\sigma_e^2} \Delta_2^2 \sum_{i=1}^k t_i(n-t_i) \text{ and } p(x; x_i^2, \eta^2)$$

denotes a noncentral chi-square density with d.f. $= f$, and a noncentrality parameter η^2 .

Notice that, if $\Delta_1 = 0$ and $\Delta_2 = 0$, then (4.5) reduces to a product of two central Wishart densities $G_{fb}(S_b, nV_\alpha + V_e) G_{fe}(S_w, V_e)$. This is a well-known result known as the multivariate Cochran's theorem (see Anderson (1958), p. 165). Similarly, $\Delta_1 = \Delta_2 = 0$ implies that (4.7) is a product of two central chi-square densities, a well-known result in the univariate random effects model.

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